

CS410 Midterm 2 Part (a)

6 December 2023

Olcay Taner YILDIZ

I. MULTIPLE CHOICE QUESTIONS (40 POINTS)

Given the following CFG C_1 . Answer questions 1-4 accordingly.

$$\begin{aligned} A &\rightarrow 0A1 \\ A &\rightarrow B \\ B &\rightarrow \# \end{aligned}$$

1. What is V ?

- a) $\{A\}$
- b) $\{A, B\}$
- c) $\{A, B, 0, 1\}$
- d) $\{0, 1, \#\}$
- e) $\{A, B, 0, 1, \#\}$

2. What is Σ ?

- a) $\{A\}$
- b) $\{A, B\}$
- c) $\{A, B, 0, 1\}$
- d) $\{0, 1, \#\}$
- e) $\{A, B, 0, 1, \#\}$

3. Which of the following string is not accepted by C_1 ?

- a) 0#1
- b) 00#11
- c) 0000#1111
- d) #
- e) All of them accepted

4. Which of the following is the correct derivation for 000#111?

- a) $A \rightarrow 0A1 \rightarrow 00B11 \rightarrow 000B111 \rightarrow 000\#111$
- b) $A \rightarrow 0A1 \rightarrow 00A11 \rightarrow 000B111 \rightarrow 000\#111$
- c) $A \rightarrow 0A1 \rightarrow 00A11 \rightarrow 000A111 \rightarrow 000\#111$
- d) $A \rightarrow 0A1 \rightarrow 00A11 \rightarrow 000A111 \rightarrow 000B111 \rightarrow 000\#111$
- e) $A \rightarrow 0A1 \rightarrow 000A111 \rightarrow 000B111 \rightarrow 000\#111$

5. Which of the following CFG generates the language $L = \{a^n c^{2n} \mid n > 0\}$?

- a) $S \rightarrow aaSc \mid \epsilon$
- b) $S \rightarrow aaSc \mid aac$
- c) $S \rightarrow aScc \mid \epsilon$
- d) $S \rightarrow aScc \mid acc$
- e) None

6. Which of the following CFG generates the language $L = \{a^{n+1}b^n \mid n \geq 1\}$?

- a) $S \rightarrow aaSb \mid \epsilon$
- b) $S \rightarrow aaSb \mid T$
 $T \rightarrow a$
- c) $S \rightarrow aSb \mid T$
 $T \rightarrow a$
- d) $S \rightarrow aSb \mid T$
 $T \rightarrow aab$
- e) None

7. Which of the following CFG generates the language $L = \{a^n b^m \mid n > m\}$?

- a) $S \rightarrow aSb \mid a$
- b) $S \rightarrow aSb \mid T$
 $T \rightarrow aa$
- c) $S \rightarrow aSb \mid T$
 $T \rightarrow aT \mid a$
- d) All of them correct
- e) None

8. What is the language generated by the following CFG?

$$S \rightarrow aaSc \mid T \mid \epsilon$$

$$T \rightarrow aaaTb \mid \epsilon$$

- a) $L = \{a^n b^m c^k \mid n, m, k \geq 0, n = 2m + 3k\}$
- b) $L = \{a^n b^m c^k \mid n, m, k \geq 0, 2n = 3m + k\}$
- c) $L = \{a^n b^m c^k \mid n, m, k \geq 0, 3n = m + 2k\}$
- d) $L = \{a^n b^m c^k \mid n, m, k \geq 0, n = 3m + 2k\}$
- e) $L = \{a^n b^m c^k \mid n, m, k \geq 0, 2n = m + 3k\}$

9. What is the language generated by the following CFG?

$$\begin{aligned} S &\rightarrow aaSbbb \mid T \\ T &\rightarrow Tb \mid \epsilon \end{aligned}$$

- a) $L = \{a^{2n}b^{3m} \mid n, m \geq 0\}$
- b) $L = \{a^{3n}b^{2m} \mid n, m \geq 0\}$
- c) $L = \{a^n b^m \mid 2m \geq 3n\}$
- d) $L = \{a^n b^m \mid 2n \geq 3m\}$
- e) $L = \{a^n b^m \mid m \geq n\}$

10. What is the language generated by the following CFG?

$$\begin{aligned} S &\rightarrow 0S1 \mid A \mid B \\ A &\rightarrow 0A \mid 0 \\ B &\rightarrow 1B \mid 1 \end{aligned}$$

- a) $L = \{0^n 1^n \mid n \geq 0\}$
- b) $L = \{0^m 1^n \mid n \geq 0, m \geq 0\}$
- c) $L = \{0^m 1^n \mid n \leq m\}$
- d) $L = \{0^m 1^n \mid m \geq n\}$
- e) $L = \{0^m 1^n \mid m \neq n\}$

11. The class of context-free languages are not closed under which of the following operation?

- a) union
- b) intersection
- c) concatenation
- d) star
- e) all

12. Given the following ambiguous CFG, which of the following strings has two different leftmost derivations?

$$\begin{aligned} S &\rightarrow 0S \mid 1A \mid 0B0 \mid 1 \\ A &\rightarrow 0A \mid \epsilon \\ B &\rightarrow 0B0 \mid 1 \end{aligned}$$

- a) 1
- b) 010
- c) 00100
- d) 0001000
- e) All

13. How many rules of the following CFG does not satisfy the rules of being in CNF?

$$\begin{aligned} S &\rightarrow 0S \mid 1A \mid 0B0 \mid 1 \\ A &\rightarrow 0A \mid \epsilon \\ B &\rightarrow 0B0 \mid 1 \end{aligned}$$

- a) 4
- b) 5
- c) 6
- d) 7
- e) 8

14. Which of the following CFG is not in CNF?

a)

$$S \rightarrow SS \mid b$$

b)

$$\begin{aligned} S &\rightarrow TT \mid a \\ T &\rightarrow b \end{aligned}$$

c)

$$\begin{aligned} S &\rightarrow a \mid ST \\ T &\rightarrow b \mid a \mid TT \end{aligned}$$

d) All of them in CNF

e) None of them in CNF

15. Which of the following CFG is in CNF?

a)

$$\begin{aligned} S &\rightarrow 0 \mid AB \mid CD \\ C &\rightarrow AB \\ D &\rightarrow AA \\ A &\rightarrow 0 \\ B &\rightarrow 1 \end{aligned}$$

b)

$$\begin{aligned} S &\rightarrow AC \mid CC \mid CE \\ E &\rightarrow DC \\ C &\rightarrow BB \\ D &\rightarrow BD \mid DA \mid a \mid b \\ A &\rightarrow a \\ B &\rightarrow b \end{aligned}$$

c)

$$\begin{aligned} S &\rightarrow AA \mid AB \mid 0 \mid 1 \\ A &\rightarrow 0 \\ B &\rightarrow 1 \end{aligned}$$

d) All of them in CNF

e) None of them in CNF

16. Given the following CFG, what is the resulting CFG after we apply unit rule removal in CNF conversion?

$$\begin{aligned} S &\rightarrow AB \mid B \\ A &\rightarrow B \\ B &\rightarrow b \end{aligned}$$

- a) $S \rightarrow AB \mid B$
 $A \rightarrow b$
 $B \rightarrow b$
- b) $S \rightarrow AB \mid b$
 $A \rightarrow b$
 $B \rightarrow b$
- c) $S \rightarrow Ab \mid b$
 $A \rightarrow b$
 $B \rightarrow b$
- d) $S \rightarrow AB \mid B$
 $A \rightarrow B$
 $B \rightarrow b$
- e) None of them

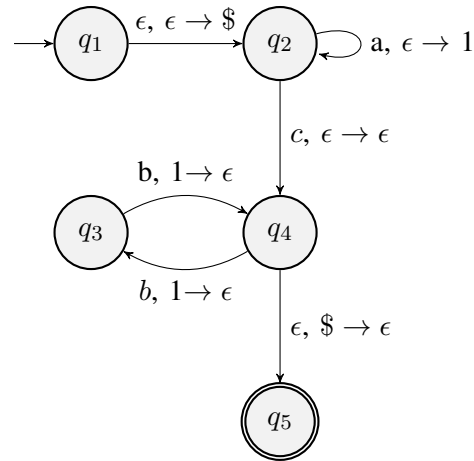
17. Which of the following languages is context-free?

- a) $L = \{a^n b^{n^2} \mid n \geq 0\}$
- b) $L = \{a^m b^n \mid n \geq 0, m \geq 0\}$
- c) $L = \{a^n b^{2n} \mid n \geq 0\}$
- d) $L = \{a^n b^n c^n \mid n \geq 0\}$
- e) All of them

18. Which of the following languages is not context-free?

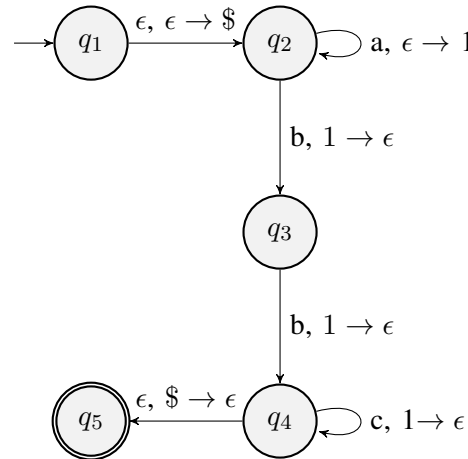
- a) $L = \{a^n b^n c^m d^m \mid n, m \geq 0\}$
- b) $L = \{a^n b^m c^n d^m \mid n, m \geq 0\}$
- c) $L = \{a^n b^m c^m d^n \mid n, m \geq 0\}$
- d) $L = \{a^m b^n c^n d^m \mid n, m \geq 0\}$
- e) All of them context-free

19. What is the language generated by the following PDA?



- a) $L = \{a^n c b^n \mid n \geq 0\}$
- b) $L = \{a^m c b^n \mid n \geq 0, m \geq 0\}$
- c) $L = \{a^n c b^{2n} \mid n \geq 0\}$
- d) $L = \{a^{2n} c b^n \mid n \geq 0\}$
- e) $L = \{a^{2n} c b^{2n} \mid n \geq 0\}$

20. What is the language generated by the following PDA?



- a) $L = \{a^{n+2} b^2 c^{2n} \mid n \geq 0\}$
- b) $L = \{a^{2n} b^2 c^{n+2} \mid n \geq 0\}$
- c) $L = \{a^{n+2} b^{2n} c^n \mid n \geq 0\}$
- d) $L = \{a^{n+2} b^2 c^n \mid n \geq 0\}$
- e) $L = \{a^n b^m c^{n+2} \mid n, m \geq 0\}$